

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460974.7966 +/- 0.0088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.057
Transit depth (Rp/Rs)^2	1.26 +/- 1.28 [%]
Orbital Inclination (inc)	89.6 +/- 2.78
Airmass coefficient 1 (a1)	1.027 +/- 0.011
Airmass coefficient 2 (a2)	-0.022 +/- 0.064
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#3 - [369, 374]
Optimal Aperture	3.99
Optimal Annulus	4.69
Transit Duration (day)	0.0283 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.112 $\pm$ 0.057 vs 0.1646 (deviation 0.92 σ)
Duration fit vs published	0.0283 d vs 0.0758 d (deviation 4.8 σ)
Mid-time O-C	18.9 min
Depth SNR	2.0
Residual scatter	1.11 %

Light curve

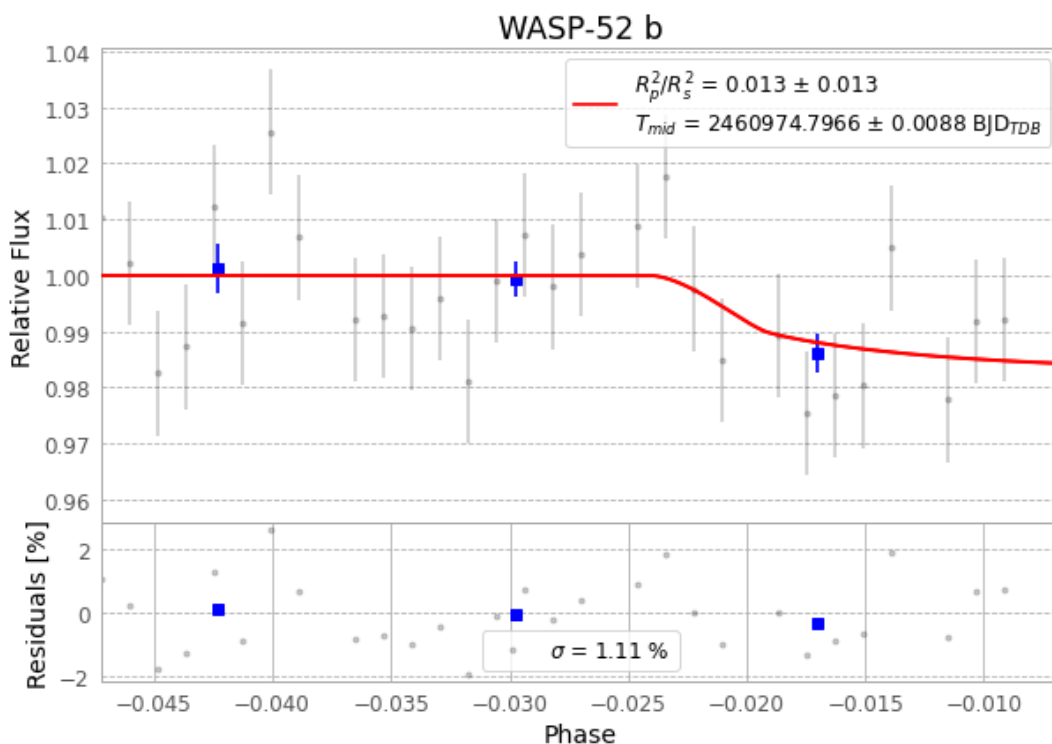


Figure 1 — FinalLightCurve_WASP-52 b_2025-10-25.png (SHA-256 44876db8782f822f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 320], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ff0a7c899823bb46...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8d12984

Data provenance

31 FITS frames (20 MB), WASP-52251026050016.FITS → WASP-52251026064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-251026.log (1029c0338e78...), FinalLightCurve_WASP-52 b_2025-10-25.png (44876db8782f...), FinalLightCurve_WASP-52 b_2025-10-25.csv (034ed653efda...)

Compute resources

Wall time 85.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 14:17Z.